



December 2025

(For BT)

Max. Marks: 100

Linear graph sheet may be supplied.

20 Marks

- 1-

9. L - amino acid oxidase acts only on L-amino acids, not on D-amino acids, here enzyme shows
- A) Group specificity
B) Stereo specificity
C) Metallo specificity
D) Geometric specificity
10. In a first-order reaction, the rate of the reaction is:
- A) Independent of the concentration of the substrate.
B) Directly proportional to the concentration of one reactant.
C) Depending on two reactants in equal amounts.
D) Independent of temperature.
11. The Lineweaver-Burk plot is a double-reciprocal plot used to determine kinetic parameters. What is plotted on the X-axis?
- A) [S] (substrate concentration)
B) $1/[S]$ (reciprocal of substrate concentration)
C) $1/V$ (reciprocal of reaction velocity)
D) V_{max} (maximum velocity)
12. Why is the Lineweaver-Burk plot considered useful in enzyme kinetics?
- A) It simplifies the determination of K_m and V_{max} by linearizing the data.
B) It provides a graphical representation of allosteric enzyme behavior.
C) It can only be used for first-order reactions.
D) It eliminates the need for experimental data collection.
13. In an Eadie-Hofstee plot, which parameters are plotted on the x-axis and y-axis, respectively?
- A) V vs. [S]
B) V vs. $V/[S]$
C) $V/[S]$ vs. V
D) [S] vs. $1/V$
14. In competitive inhibition, the inhibitor:
- A) Binds to the enzyme-substrate complex, preventing the reaction.
B) Binds to the active site of the enzyme, competing with the substrate.
C) Alters the enzyme's structure permanently.
D) Binds to an allosteric site, increasing enzyme activity.
15. What is the effect of low temperature on enzyme activity?
- A) Enzymes work best at any temperature
B) Lower temperature always enhances enzyme activity.
C) Enzyme shows decreased activity.
D) Low temperature denatures enzyme
16. What is the primary goal of screening for novel biocatalysts?
- A) To identify the most common microorganisms in each environment.
B) To discover new enzymes that can catalyze specific biochemical reactions.
C) To increase the yield of existing enzyme production methods.
D) To characterize the metabolic pathways of known microorganisms.
17. Which technique is suitable to separate enzymes having same molecular size?
- A) Size exclusion chromatography
B) Isoelectric focusing
C) SDS-PAGE
D) Ultracentrifugation
18. Which of the following salt is used for salting out of enzyme protein?
- A) Sodium Sulphate
B) Ammonium Sulphate
C) Copper Sulphate
D) Barium Sulphate
19. In encapsulation for enzyme immobilization, what is the primary feature of this method?
- A) The enzyme is physically trapped in a matrix.
B) The enzyme is simply mixed with a substrate.
C) The enzyme forms strong covalent bonds with a support material.
D) The enzyme remains soluble in the reaction mixture.
20. _____ will be useful if there is utilization or release of gaseous material
- A) CONTINUOUS reactor
B) Fluidized – Bed Reactor
C) Batch membrane reactor
D) Batch reactor

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Discuss classification of enzymes as per Enzyme Commission with suitable example.	08	L*2	1	1
b) Narrate the historical development of enzyme technology from early observations to modern understanding.	08	L1	1	1
2. a) Which theory explains that 'substrate is strained due to the induced conformation change in the enzyme'? Comment with example.	08	L1	1	1
b) Elaborate on allosteric enzymes with their properties and applications.	08	L2	1	1

Unit – II

3. a) With the help of an example, describe the mechanism of covalent catalysis in enzymatic reactions.	08	L2	2	2,3
b) What is sigmoidal kinetics? Explain Monod-Wyman-Changeux and Koshland-Némethy-Filmer models of enzyme kinetics.	08	L2	3	2,3
4. a) Derive Michaelis-Menten equation and add a note on applications of K_m	08	L2	3	2,3
b) How to determine optimum temperature for a given enzyme? Why it is specific?	08	L1	2	2,3
5. a) Assume an enzyme kinetics experiment that gives the following initial reaction velocities (V_o) at different substrate concentrations ($[S]$): calculate K_m using obtained data.				

[S] (mM)	V_o ($\mu\text{mol/min}$)
1	10
5	40
10	70
15	85
20	95

b) Give a brief account on Non-competitive and Uncompetitive enzyme inhibition with graphical representation.	08	L1	3	2,3
	08	L2	2	2,3

Unit – III

6. a) Give a general procedure for isolation and selection of microbes involved in enzyme production.	08	L1	4	3,4
b) Discuss various methods of High throughput screening for enzyme.	08	L1	5	2
7. a) Comment on criteria of purity for a commercially important enzyme.	08	L1	4	3,4
b) Immobilization plays a significant role in enzyme technology. Justify.	08	L2	5	2
8. a) Give a general diagram of fluidized bed bioreactors for immobilized enzyme-based reaction. Add a note on its parameters and significance.	08	L2	5	2
b) What is tailor made biocatalyst? Discuss the methods involved in its preparation.	08	L2	4	3,4

- BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
